# Jiaqi Ge

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## Education

**Guangdong Technion – Israel Institute of Technology** Sep 2023 – Present

*B.Sc. in Mathematics and Applied Mathematics* *Shantou, China*

- **Relevant coursework:** Probability Theory, Infinitesimal Calculus I–III, Algebra A & B, Introduction to Numerical Analysis, Introduction to Applied Mathematics, Combinatorics.

**Technion – Israel Institute of Technology** Summer 2026

*Summer Research Exchange Student; Non-Euclidean Deep Learning* *Haifa, Israel*

## Publications

Haotian Lin, Silin Chen, Xiaodong Gu, Yuling Shi, Chengxi Pan, **Jiaqi Ge**, Mengfan Li, Jianghong Huang, Mengchieh Chuang, Beijun Shen, and Haibing Guan.
**"Know Before Fix: QA-Driven Repository Knowledge Acquisition for Software Issue Resolution."** *arXiv preprint*, arXiv:2607.11111, 2026.

## Research Experience

**Research Assistant – Mixture-of-Experts Pruning** Jun 2026 – Present

*Mohamed bin Zayed University of Artificial Intelligence; Advisor: Prof. Lijie Hu*

- Investigated ACDC-style iterative pruning of mixture-of-experts (MoE) models, using forward passes on the *currently pruned* model to re-evaluate candidate expert removals without downstream-task labels or post-pruning fine-tuning.
- Explored router interactions, groupwise forward verification, and candidate-group splitting. Designed 25% expert-pruning comparisons on LFM2.5-8B-A1B, OLMoE-1B-7B, Mixtral-8x7B-v0.1, Qwen3-30B-A3B, GLM-4.7-Flash, and Gemma-4-26B-A4B against Random, Frequency, EAN, and official REAP, using BoolQ/ARC/MMLU accuracy and C4/WikiText-2 perplexity.
- **Current results at 25% pruning (vs. official REAP):** A REAP-ranked, forward-verified variant improved LFM2.5-8B-A1B four-task macro accuracy from 62.22% to **65.93%** (+3.71 points) and C4 perplexity from 16.465 to **15.601**; a *distinct basic forward-verified variant* improved OLMoE-1B-7B macro accuracy from 41.44% to **51.67%** (+10.23 points) and C4 perplexity from 15.184 to **14.290**. Further evaluations on the remaining MoE models and at 50% pruning are ongoing.
- *First-author manuscript in preparation for submission to an ICLR workshop.*

**Research Assistant – Repository QA for Software Issue Resolution** Feb 2026 – May 2026

*Shanghai Jiao Tong University; Advisor: Prof. Xiaodong Gu*

- Designed baseline-versus-QA-augmented experiments for *ACQUIRE* on SWE-bench Verified (500 issues) and built an LLM-as-a-judge pipeline for question-level, set-level, and pairwise evaluation. The coauthored study improved Pass@1 from 58.4% to **62.2%** (+3.8 points) on GPT-5-mini and from 66.4% to **70.8%** (+4.4 points) on DeepSeek-V3.2.
- Conducted step-level trajectory analysis of 44 Fail-to-Pass cases under DeepSeek-V3.2, segmenting repair actions into Locating, Reproducing, Fixing, and Verifying and examining QA-grounded actions and patch alignment. QA injection reduced agent rounds by **17.1%** and Locating/Fixing steps by **23.8%**/**24.1%**; QA-related steps comprised **51.5%** of Locating and **50.6%** of Fixing actions.
- Analyzed ablations showing that removing question decomposition and category-guided generation lowered Pass@1 by **4.8**/**3.8** points, respectively; contributed to *Know Before Fix* through figures, related-work synthesis, experimental interpretation, writing, and revision.

**Research Exchange – Fourier Sliced-Wasserstein (FSW) Document Classification** Summer 2026

*Technion – Israel Institute of Technology; Supervisor: Tal Amir* *Haifa, Israel*

- Independently implemented a document-classification pipeline that maps probability distributions over 300-dimensional fastText vectors into Euclidean space using the existing Fourier Sliced-Wasserstein (FSW) embedding, replacing exact Word Mover's Distance (WMD) with Euclidean $k$NN search.
- Benchmarked FSW against exact WMD on Amazon, Classic, and Reuters: obtained approximately **150**× faster per-query classification (∼0.02 vs. ∼3 s), with **1.5–2.6**× higher test error. Designed and implemented FSW shortlist-and-exact-WMD reranking to recover near-WMD accuracy with **estimated 4–29**× speedups.
- Implemented $k$-means and spectral clustering and trained cross-entropy/contrastive MLP models on FSW vectors; the best models beat exact-WMD $k$NN test error on Amazon and Reuters.

## Projects

**Grammar-Based Fuzzing for Metamorphic Testing of NLP Models** Oct 2025 – Feb 2026

*Guangdong Technion – Israel Institute of Technology; Advisor: Prof. Nazareno Aguirre*

- Designed grammar-based fuzzing for both multi-step text-perturbation pipelines and metamorphic-property generation; integrated the resulting `INV/DIR` tests with CheckList to evaluate NLP sentiment models beyond single-step edits.
- Applied fuzzed transformation chains to 100 Twitter airline tweets using a Hugging Face classifier; a nine-step case study exposed counterintuitive label flips and directional-expectation violations under interacting negation and formatting edits. Examined semantic drift as a threat to property validity. [Report] [Code]

**Numerical Boundary Value Problems with Chebfun** Aug 2025 – Nov 2025

*Guangdong Technion – Israel Institute of Technology; Advisor: Prof. Antti Rasila*

- Reviewed initial- and boundary-value problems, organizing existence/uniqueness theorems, Green's functions, and Sturm–Liouville theory in a 20-page report. [Report]
- Implemented MATLAB Chebfun examples for nonlinear BVPs, singular eigenvalue problems, and dynamical systems; on an exactly solvable benchmark, achieved $7.9 \times 10^{-14}$ maximum error with 13 degrees of freedom, versus $4.4 \times 10^{-7}$ for 1,024-point second-order finite differences. [Code]

## Work Experience

**Undergraduate TA – Linear Algebra and Calculus** Mar–Aug 2026; Oct 2026–Present

*Guangdong Technion – Israel Institute of Technology* *Shantou, China*

- Support weekly workshops for classes of approximately 60 students, helping them work through mathematical problems and clarify concepts in linear algebra and calculus.

**Undergraduate Teaching Assistant – Python** Oct 2025 – Jan 2026

*Guangdong Technion – Israel Institute of Technology* *Shantou, China*

- Supported introductory programming labs and helped students debug code and understand Python fundamentals.

**Homework Grader – Differential and Integral Calculus 1M** Oct 2026 – Present

*Guangdong Technion – Israel Institute of Technology* *Shantou, China*

- Grade assignments and provide written feedback on the mathematical reasoning in students' solutions.

## Awards

**Chancellor's Scholarship** – US$28,500 total 2023 – 2027

- Awarded by Guangdong Technion – Israel Institute of Technology for outstanding performance in the Gaokao.

## Skills

**Programming:** Python, C, C++, Java, MATLAB, Assembly

**Tools & Libraries:** PyTorch, Hugging Face Transformers, Git, Linux, MATLAB Chebfun, CheckList, Fuzzing Book (GrammarFuzzer)

**Languages:** Chinese (native), English (fluent)